

INTRODUCTION

Libraries remain an essential part of academic institutions as they provide access to learning resources and quiet study spaces. However, many schools and universities experience overcrowding in their libraries due to increasing student population and limited physical space. This situation affects students' comfort, concentration, and overall learning experience.

To address these issues, modern libraries are gradually

adopting digital solutions such as Smart Library Management Systems (SLMS), which include features like real-time occupancy monitoring and seat reservation. These technologies aim to improve space utilization and reduce overcrowding inside libraries.

BODY

Library Overcrowding and Space Management

Library overcrowding has become a common issue in

academic institutions where student demand exceeds available seating capacity. According to Tenopir (2012), the increasing use of academic libraries reflects the growing need for study spaces, especially during peak academic periods.

Similarly, the International Federation of Library Associations and Institutions (2021) emphasized that libraries must ensure safe and efficient space management to accommodate growing user populations. Proper monitoring of library occupancy

is essential to maintain order and user satisfaction.

Smart Library Systems and Digital Solutions

The American Library Association (2022) stated that libraries are now transforming into hybrid learning environments that combine physical and digital services. Because of this, many institutions are adopting smart systems to improve efficiency and user experience.

In addition, studies on Internet of Things (IoT) show that real-time monitoring systems can track user presence and improve space management in public environments. These technologies are also being applied in libraries to reduce overcrowding and improve accessibility.

Seat Reservation and Occupancy Monitoring Systems

Recent studies have shown that seat reservation systems help reduce waiting time and

improve user satisfaction in libraries. According to Patel (2023), smart library systems with reservation features allow users to secure seats before entering the facility, preventing unnecessary crowding.

Zhang et al. (2022) also found that IoT-based occupancy monitoring systems can effectively track the number of users in real time, allowing administrators to manage space more efficiently.

CHALLENGES

1. Limited Physical Space

Many academic libraries struggle with limited seating capacity, which leads to overcrowding during peak hours (Tenopir, 2012).

2. Inefficient Manual Monitoring

Traditional monitoring methods are often manual and inaccurate, making it difficult to track the actual number of users inside the library (IFLA, 2021).

3. Lack of Real-Time Information

Students often do not know if seats are available before going to the library, which contributes to overcrowding issues.

4. System Accessibility Issues

Some digital systems require stable internet connections, which may affect real-time monitoring performance.

RESEARCH GAPS

Based on the reviewed literature, the following gaps were identified:

Lack of Full Integration

Most existing systems focus only on either book management or attendance tracking, but not both combined with space management.

Limited Real-Time Seat Visibility

There are few systems that allow students to see real-time seat availability before entering the library.

Focus on Book Systems, Not Crowd Control

Most library systems are designed for cataloging and borrowing, not for managing overcrowding inside the library.

CONCLUSION

The reviewed literature shows that library overcrowding is still a major issue in academic institutions. While existing systems have improved library operations, most of them do not fully address space management and real-time occupancy control.

Therefore, there is a need for a Smart Library Management System that focuses on overcrowding solutions through seat reservation and real-time monitoring. This study aims to fill that gap by providing a more efficient and user-centered library management system.

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